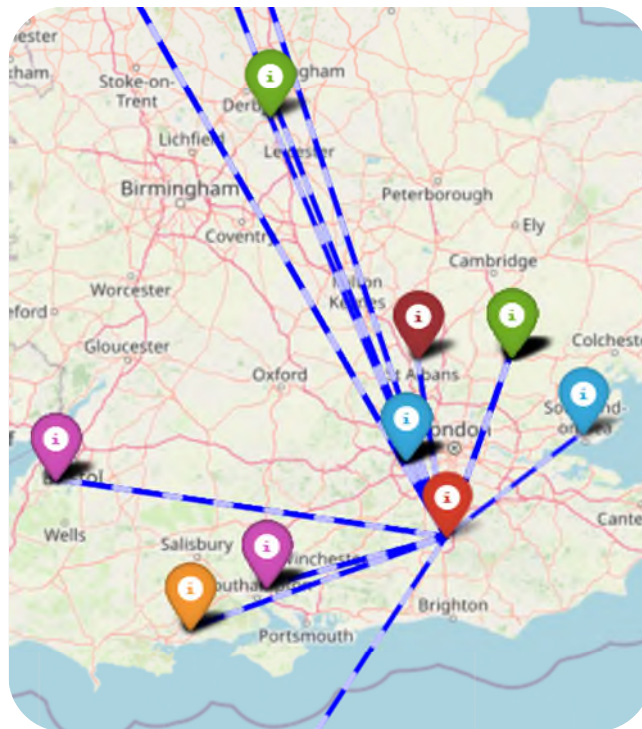


MARS

Modelling Aviation Resilience Scenarios

for the UK Aviation System



Dr Fabian Steinmann



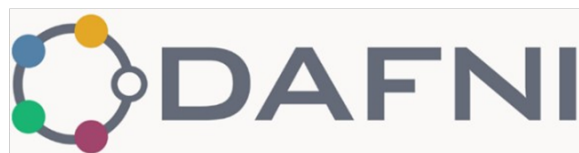
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Background

Mass diversion events present a major challenge to the UK aviation system. These events place significantly operational pressure on the network and substantially increase the workload for the air traffic control officers, who must quickly identify multiple suitable diversion airports. This process can be quite time-consuming, which could lead to multiple fuel emergencies, thereby compromising the stability of the aviation system. An important contributing factor is limited spare capacity in the system, a consequence of high utilization at the major UK airports as highlighted by a 2017 study conducted by the Civil Aviation Authority.

Objectives

The Mass Diversion Simulation Tool (MassDivST) simulates airport closures within the UK aviation system and models the subsequent diversion process. Using pre-defined input criteria, an algorithm calculates optimal diversion airports for affected flights without compromising the operation at these alternative aerodromes. The outputs can be used to identify potential bottlenecks and support discussions on creating additional buffer capacity in the UK aviation system, such as increasing pre-approved diversion slots at airports.

Algorithm Design

A digital model of the 34 largest UK commercial airports have been designed and MassDivST uses historical data on arrival and departures at those airports. The user can simulate the closure of a selected airport for different time periods to observe the diversion dynamics.

Key Parameters for Diversion Decision

The algorithm identifies the most suitable diversion airport for each aircraft scheduled to arrive during the closure period. It evaluates alternate airports based on the following parameters:

- Declared diversion priorities
- Available stands at diversion airports
- Distance to nearest airport
- Contractual agreements

Declared Diversion Priorities

The tool considers up to two declared diversion airports per flight. However, due to multiple aircraft designating the same diversion airports in their flight plans and the capacity constraints at those locations, alternative airports are often necessary.

Available stands at diversion airports

All aircraft types have been converted to aircraft categories based on their wingspan (see Table 1).

Table 1 Aircraft categories

Code letter	Wingspan
B	15 – 24 m
C	24 – 36 m
D	36 – 52 m
E	52 – 65 m
F	65 – 80 m

The initial number of available stands at each airport can be adjusted within the simulation, enabling the modelling of various scenarios. For instance, Airport A might have the following stand availability at the start of the simulation as shown in Table 2.

Table 2 Example for available aircraft stands at airport

Airport A				
B	C	D	E	F
0	8	3	1	1

The stand availability is dynamically updated during simulations as flights arrive and depart. It is important to note that smaller aircraft can use larger stands, but the reverse is not possible.

Distance to nearest airport

Using GPS data, MassDivST calculates the distance to the nearest diversion airport, estimates the additional flight time, and checks if a stand is available at the calculated arrival time.

Contractual agreements

Agreements, for example, between airlines and ground handlers further restrict diversion options. A compatibility matrix within the dataset ensures compliance with these agreements (see Table 3).

Table 3 Example for contractual agreements at airports

	Airport A	Airport B	Airport C	Airport D	...
Airline 1	Yes	Yes	No	No	
Airline 2	No	Yes	Yes	Yes	
Airline 3	No	No	Yes	No	
...					

Simulation Outputs

The MassDivST generates several post simulation reports.

Overview of Diversion Airports

The tool will provide an overview of the total number of diversions during the selected closure period, along with a breakdown of the airports to which the aircraft were diverted. Figure 1 shows the results of a three-hour closure at London Gatwick from 12:00 – 15:00 UTC.

Additionally, a map of UK airports is generated, providing a visual reference of the diversions. Users can click on individual airports to view the number of aircraft diverted to each location (Figure 2).

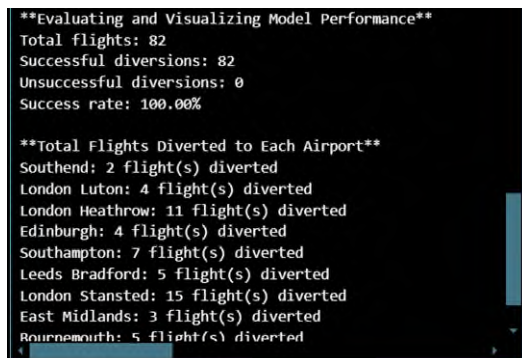


Figure 1 Breakdown of diverted flights by alternate airport

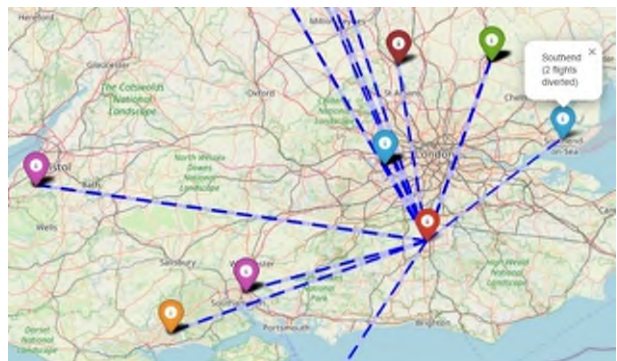


Figure 2 Interactive map displaying diversion airports

A detailed list of all diverted flights is also provided, showing their corresponding diversion airports and updated arrival times based on the distance to the alternative airport (Figure 3).

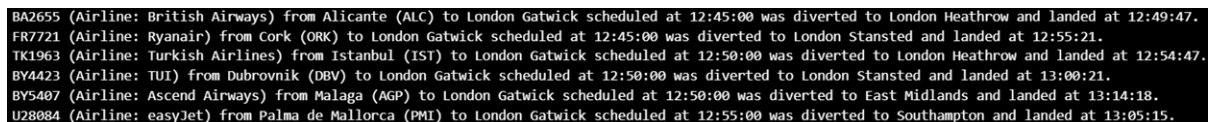


Figure 3 Flight-by-flight list showing diversion destinations and updated arrival times

Get in touch

For inquiries or collaboration opportunities, please contact: f.steinmann@cranfield.ac.uk.